How Smartphones Work

Introduction

A smartphone is a handheld computer that combines the functions of a mobile phone with those of a computer, camera, music player, GPS navigator and much more. Unlike older feature phones, smartphones run a full operating system, such as Google's Android or Apple's iOS, and let users install apps. The first handheld mobile phone call was made by Martin Cooper of Motorola on 3 April 1973. The IBM Simon, launched in 1994, is considered the first smartphone, and the first iPhone, launched by Apple in 2007, and the first Android phone, in 2008, started the modern smartphone era. More than 6 billion people worldwide now use smartphones, and India, with over 700 million users, is the second-largest smartphone market in the world.

Main Hardware Components

A smartphone has several key parts: the processor (SoC), memory (RAM), storage, display and touchscreen, battery, cameras, sensors, wireless modems and the operating system software that ties it all together. A printed circuit board holds the chips, which are protected by a metal or glass body.

What Is a SoC?

SoC stands for System on a Chip. It is a single integrated circuit that contains most of the electronic parts of a computer on one tiny piece of silicon, instead of using many separate chips. A smartphone SoC combines the CPU (central processing unit, the main "brain" with several cores, usually using the ARM architecture, for running apps), the GPU (graphics processing unit, for games and graphics), the NPU or AI engine (for machine learning tasks such as face recognition and camera effects), the ISP (image signal processor, which processes camera images), the memory controller, the modem for 4G and 5G cellular connectivity, support for Wi-Fi, Bluetooth and GPS, and a video encoder and decoder. Combining everything in one chip makes the phone smaller, faster and more power efficient. Well-known smartphone SoCs include Qualcomm Snapdragon, Apple's A-series chips (such as the A17 Pro), MediaTek Dimensity, Samsung Exynos and Google Tensor. They are made at advanced chip factories, such as TSMC in Taiwan, using process nodes of 3 to 5 nanometres, with billions of transistors. RAM, the working memory (commonly 6 to 16 GB), is separate but is often placed close to the SoC, and storage uses fast flash memory, called UFS, with 128 GB to 1 TB.

How Does a Touchscreen Detect Touch?

Most smartphones use a capacitive touchscreen. The glass screen is coated with a thin transparent layer of a conductive material, usually indium tin oxide, arranged as a grid of tiny electrodes in rows and columns, which creates a weak electrostatic field over the screen. The human body is a conductor of electricity. When a finger touches or comes near the glass, it draws a tiny amount of electric charge from that spot and changes the capacitance, the ability to store charge, at that location. The touch controller chip continuously measures the capacitance at every point of the grid, detects where the change occurred, and calculates the exact coordinates of the touch, then sends this to the processor. Because the grid can detect many points simultaneously, it supports multi-touch gestures such as pinch to zoom and swipe, and the system responds within milliseconds. This is why ordinary gloves and plastic styluses do not work on a capacitive screen, unless they are conductive. Older resistive touchscreens, in which pressure pushes two layers together, worked with any object but were less sensitive and did not support multi-touch easily. The display itself is usually either an LCD, which uses a backlight with liquid crystals, or an OLED or AMOLED, in which each pixel gives off its own light, giving deeper blacks and thinner screens. Refresh rates of 90 or 120 Hz make scrolling smooth. Gorilla Glass or similar tough glass protects the screen.

What Type of Battery Do Smartphones Use?

Smartphones use lithium-ion (Li-ion) or lithium-polymer (Li-Po) batteries, which are rechargeable. They are favoured because of their high energy density, which means they store a lot of energy in a small, light package, their ability to be recharged hundreds of times, and their low self-discharge. A typical battery has a capacity of about 3,000 to 6,000 milliamp-hours (mAh) at around 3.7 to 3.85 volts. Batteries are charged using a USB-C port or wirelessly using the Qi standard. Fast chargers, from 20 watts to over 100 watts, charge a battery to 50 percent in about 20 to 30 minutes. A battery gradually loses capacity, usually degrading to around 80 percent after 500 to 1,000 full charge cycles, and heat speeds up ageing. Battery safety circuits protect against overcharging and overheating. Tips: avoid extreme heat, don't leave the phone at 0 percent or 100 percent for long, and use certified chargers. Newer silicon-carbon batteries offer higher capacity. John Goodenough, Stanley Whittingham and Akira Yoshino won the 2019 Nobel Prize in Chemistry for lithium-ion batteries.

Sensors, Cameras and Connectivity

Smartphones contain many sensors: an accelerometer measures movement and tilt, a gyroscope measures rotation, a magnetometer works as a digital compass, a proximity sensor turns off the screen during a call, an ambient light sensor adjusts brightness, a barometer measures air pressure, a fingerprint sensor and a face recognition system provide security, and GPS and NavIC receivers give location. Cameras use CMOS image sensors with lenses, often with multiple lenses (wide, ultra-wide, telephoto), optical image stabilisation and computational photography, in which software merges several images to improve quality, as in night mode. Connectivity includes 4G LTE and 5G cellular modems, Wi-Fi, Bluetooth, NFC for contactless payments, and a SIM card or eSIM. When you make a call, the phone's modem and antenna send radio waves to the nearest cell tower, and the network connects you to the other person.

The Operating System and Apps

The operating system (Android, which is based on Linux, or iOS) manages the hardware, memory, security and apps, and provides the user interface. Apps are downloaded from stores, such as Google Play and the App Store, and use hardware such as the camera and GPS with the user's permission. Regular software updates add features and security fixes.

Smartphones in India and Tamil Nadu

India is the world's second-largest manufacturer of mobile phones. The Samsung factory at Noida in Uttar Pradesh, opened in 2018, is among the largest mobile phone factories in the world. Tamil Nadu is a major electronics hub: the Sriperumbudur belt near Chennai hosts the iPhone assembly plants of Foxconn and other electronics factories, and smartphones are a leading item in India's exports. The government's Production Linked Incentive (PLI) scheme, introduced in 2020, supports local manufacturing. Cheap data and UPI have made smartphones central to daily life in India.

Challenges and Future

Challenges include battery life, heat, electronic waste (e-waste), privacy, addiction and repairability. Future smartphones will have more powerful on-device AI, foldable screens, satellite connectivity, better batteries and 6G.

Quick Facts

- A SoC (System on a Chip) integrates the CPU, GPU, modem and other parts on a single chip.
- Most smartphones use capacitive touchscreens, which detect the change in electrical charge caused by a finger.
- Smartphones use lithium-ion or lithium-polymer batteries.

- The first handheld mobile phone call was made on 3 April 1973 by Martin Cooper.
- Foxconn assembles iPhones at Sriperumbudur, Tamil Nadu.